

DATA: 21.01.2021

GRUPA 403A, AIA

ANUL I

LOGICĂ COMPUTAȚIONALĂ
NR. BILET 8

$$\begin{aligned}
 1. \quad a) \quad 110110.101_{(2)} &= 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 0 \cdot 2^0 + 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} \\
 &= 32 + 16 + 0 + 4 + 2 + 0 + 0,5 + 0 + 0,125 \\
 &= 54 + 0,625 \\
 &= 54,625_{(10)}
 \end{aligned}$$

$$54,625_{(10)} = 110110.101_{(2)}$$

$$\begin{array}{l}
 54 : 2 = 27 \text{ r } 0 \\
 27 : 2 = 13 \text{ r } 1 \\
 13 : 2 = 6 \text{ r } 1 \\
 6 : 2 = 3 \text{ r } 0 \\
 3 : 2 = 1 \text{ r } 1 \\
 1 : 2 = 0 \text{ r } 1
 \end{array}
 \quad \uparrow$$

$$\begin{array}{l}
 0,625 \times 2 = 1 + 0,250 \Rightarrow a_{-1} = 1 \\
 0,250 \times 2 = 0 + 0,500 \Rightarrow a_{-2} = 0 \\
 0,500 \times 2 = 1 + 0,000 \Rightarrow a_{-3} = 1
 \end{array}
 \quad \downarrow$$

$$0,625_{(10)} = 101$$

$$54_{(10)} = 110110_{(2)}$$

b) prin mărime n semn (cod direct)

$$[+34]_{\text{dir}} = 0100010 \quad [-34]_{\text{dir}} = 1100010$$

$$\begin{array}{l}
 34 : 2 = 17 \text{ r } 0 \\
 17 : 2 = 8 \text{ r } 1 \\
 8 : 2 = 4 \text{ r } 0 \\
 4 : 2 = 2 \text{ r } 0 \\
 2 : 2 = 1 \text{ r } 0 \\
 1 : 2 = 0 \text{ r } 1
 \end{array}
 \quad \uparrow$$

prin complement față de 2 (cod complementat)

$$[+34]_{\text{comp}} = 0100010$$

$$\begin{array}{r}
 1011101 \\
 +1 \\
 \hline
 1011110 \Rightarrow [-34]_{\text{comp}} = 1011110
 \end{array}$$

per complement față de 1 (cod invers)

$$[+34]_{\text{inv}} = 0100010$$

$\Downarrow$

$$1011101 \Rightarrow [-34]_{\text{inv}} = 1011101$$

2. $x_1 \bar{x}_2 x_3 + \bar{x}_2 (\bar{x}_1 \bar{x}_2 x_3 + \bar{x}_4) = \bar{x}_2 x_3 + \bar{x}_2 \bar{x}_4$

x_1	x_2	x_3	x_4	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}_3$	$\bar{x}_4$	$x_1 \bar{x}_2 x_3$	$\bar{x}_2 (\bar{x}_1 \bar{x}_2 x_3 + \bar{x}_4)$	$x_1 \bar{x}_2 x_3 + \bar{x}_2 (\bar{x}_1 \bar{x}_2 x_3 + \bar{x}_4)$
1	1	1	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	1	0	0	0
1	1	0	1	0	0	1	0	0	0	0
1	1	0	0	0	0	1	1	0	0	0
1	0	1	1	0	1	0	0	1	0	1
1	0	1	0	0	1	0	1	1	1	1
1	0	0	1	0	1	1	0	0	0	0
1	0	0	0	0	1	1	1	0	1	1
0	1	1	1	1	0	0	0	0	0	0
0	1	1	0	1	0	0	1	0	0	0
0	1	0	1	1	0	1	0	0	0	0
0	1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	1	0	0	0	1	1
0	0	1	0	1	1	0	1	0	1	1
0	0	0	1	1	1	1	0	0	0	0
0	0	0	0	1	1	1	1	0	1	1

x_1	x_2	x_3	x_4	$\bar{x}_1$	$\bar{x}_2$	$\bar{x}_3$	$\bar{x}_4$	$\bar{x}_2 x_3$	$\bar{x}_2 \bar{x}_4$	$\bar{x}_2 x_3 + \bar{x}_2 \bar{x}_4$
1	1	1	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	1	0	0	0
1	1	0	1	0	0	1	0	0	0	0
1	1	0	0	0	0	1	1	0	0	0
1	0	1	1	0	1	0	0	1	0	1
1	0	1	0	0	1	0	1	1	1	1
1	0	0	1	0	1	1	0	0	0	0
1	0	0	0	0	1	1	1	0	1	1
0	1	1	1	1	0	0	0	0	0	0
0	1	1	0	1	0	0	1	0	0	0
0	1	0	1	1	0	1	0	0	0	0
0	1	0	0	1	0	1	1	0	0	0
0	0	1	1	1	1	0	0	1	0	1
0	0	1	0	1	1	0	1	1	1	1
0	0	0	1	1	1	1	0	0	0	0
0	0	0	0	1	1	1	1	0	1	1

Conform tabelelor de adevăr, identitatea dată este adevărată.

$$3. \quad \overline{x_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 x_3}$$

$$= \overline{x_1 x_2 \bar{x}_3} \cdot \overline{x_1 \bar{x}_2 x_3}$$

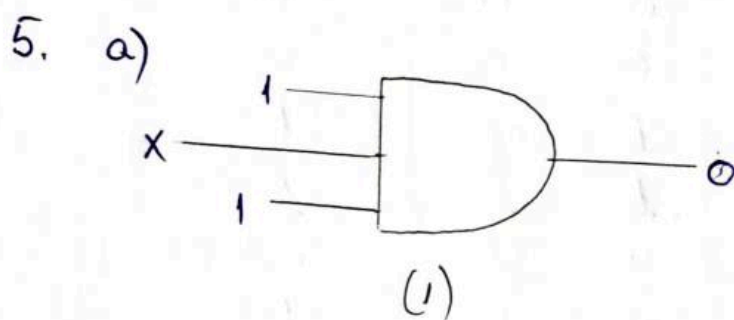
$$= x_1 x_2 \bar{x}_3 \cdot (\bar{x}_1 + x_2 + \bar{x}_3)$$

$$= \overset{0}{x_1 \bar{x}_1} x_2 \bar{x}_3 + x_1 x_2 x_2 \bar{x}_3 + x_1 x_2 \bar{x}_3 \bar{x}_3$$

$$= 0 + x_1 x_2 \bar{x}_3 + x_1 x_2 \bar{x}_3$$

$$= x_1 x_2 \bar{x}_3$$

$$\begin{aligned}
 4. \quad f(x_1, x_2, x_3) &= x_1 \bar{x}_3 + x_2 + \bar{x}_1 \bar{x}_2 \bar{x}_3 \\
 &= x_1 \bar{x}_3 (x_2 + \bar{x}_2) + x_2 (x_1 + \bar{x}_1) + \bar{x}_1 \bar{x}_2 \bar{x}_3 \\
 &= x_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 + \bar{x}_1 x_2 + \bar{x}_1 \bar{x}_2 \bar{x}_3 \\
 &= x_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 (x_3 + \bar{x}_3) + \bar{x}_1 x_2 (x_3 + \bar{x}_3) + \bar{x}_1 \bar{x}_2 \bar{x}_3 \\
 &= \underline{x_1 x_2 \bar{x}_3} + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 x_3 + \underline{x_1 x_2 \bar{x}_3} + \bar{x}_1 x_2 x_3 + \bar{x}_1 x_2 \bar{x}_3 + \bar{x}_1 \bar{x}_2 x_3 \\
 &= x_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 x_3 + \bar{x}_1 x_2 x_3 + \bar{x}_1 x_2 \bar{x}_3 + \bar{x}_1 \bar{x}_2 x_3 \\
 &= P_{10} + P_8 + P_{11} + P_3 + P_2 + P_0 \\
 &= \sum(0, 2, 3, 8, 10, 11)
 \end{aligned}$$



Poarta logică și (AND) cu trei intrări

intrări: 1, x, 1

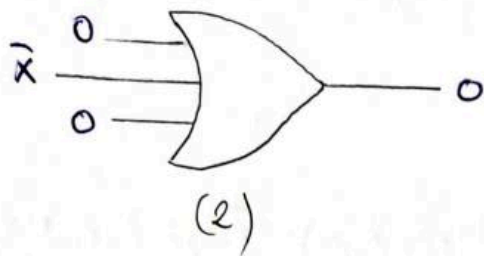
ieșiri: 0

$$1 \cdot x \cdot 1 = 0$$

$$x = 0$$

Valoarea pe care o poate lua x este 0

a	b	c	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



Poarta logică SAU (OR) cu trei intrări

intrări: 0, $\bar{x}$, 0

ieșiri: 0

$$0 + \bar{x} + 0 = 0$$

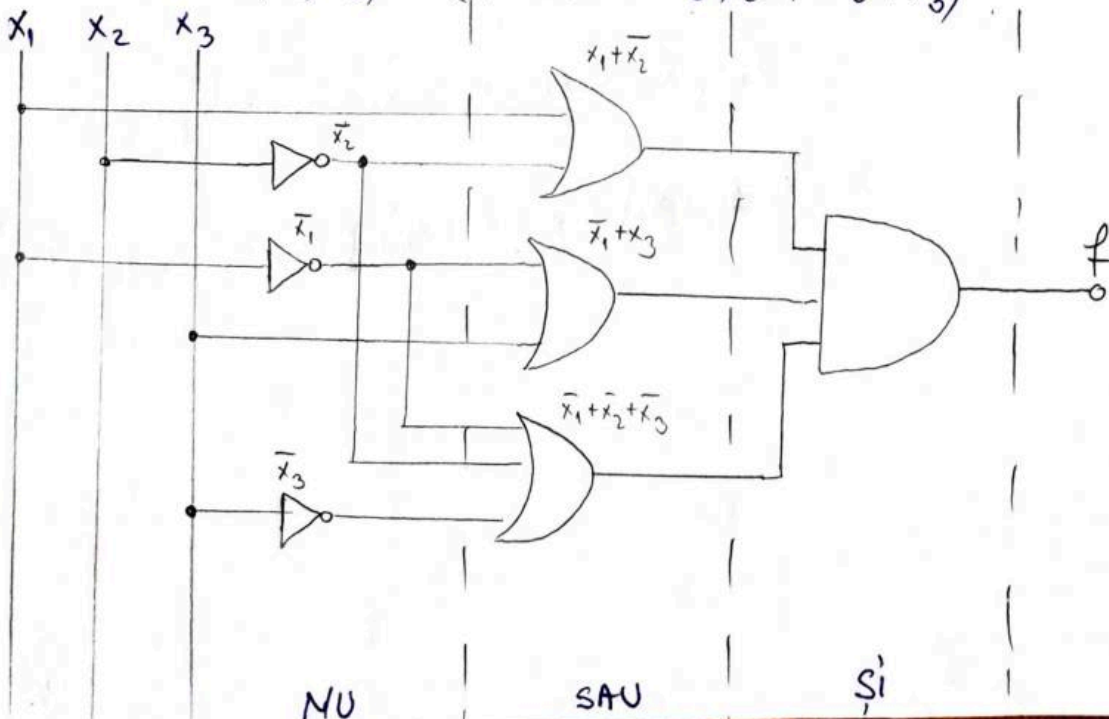
$$\bar{x} = 0 \Rightarrow x = 1$$

Valoarea pe care o poate lua x este 1

→

a	b	c	f
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

b) $f^{TMC}(x_1, x_2, x_3) = (x_1 + \bar{x}_2)(\bar{x}_1 + x_3)(\bar{x}_1 + \bar{x}_2 + \bar{x}_3)$



6. (a) $f^{FAD}(x_1, x_2, x_3, x_4) = \sum (1, 4, 5, 6, 7, 9, 13)$

Diagrama Karnaugh

$x_1 x_2$ $x_3 x_4$	00	01	11	10
00	0	1	0	0
01	1	1	1	1
11	0	1	0	0
10	0	1	1	0

Groupings:
 $\bar{x}_1 \bar{x}_2$ (top row, column 01)
 $\bar{x}_3 x_4$ (middle row, columns 00, 01, 11, 10)
 $x_2 x_3 \bar{x}_4$ (bottom row, columns 01, 11, 10)

$$f^{FAD}(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_2 + \bar{x}_3 x_4 + x_2 x_3 \bar{x}_4$$

$x_1 x_2$ $x_3 x_4$	00	01	11	10
00	0	1	0	0
01	1	1	1	1
11	0	1	0	0
10	0	1	1	0

Groupings:
 $\bar{x}_2 \bar{x}_4$ (top row, columns 01, 11)
 $x_1 \bar{x}_3 \bar{x}_4$ (middle row, columns 00, 01, 11, 10)
 $x_1 x_3 x_4$ (bottom row, columns 01, 11, 10)
 $\bar{x}_2 x_3$ (left column, rows 01, 11, 10)

$$f^{FAD}(x_1, x_2, x_3, x_4) = \bar{x}_2 \bar{x}_4 + x_1 \bar{x}_3 \bar{x}_4 + x_1 x_3 x_4 + \bar{x}_2 x_3$$

$$f^{FAD}(x_1, x_2, x_3, x_4) = \bar{x}_2 \bar{x}_4 + x_1 \bar{x}_3 \bar{x}_4 + x_1 x_3 x_4 + \bar{x}_2 x_3$$

$$= \overline{\bar{x}_2 \bar{x}_4} \cdot \overline{x_1 \bar{x}_3 \bar{x}_4} \cdot \overline{x_1 x_3 x_4} \cdot \overline{\bar{x}_2 x_3}$$

$$= (x_2 + x_4) (\bar{x}_1 + x_3 + x_4) (\bar{x}_1 + \bar{x}_3 + \bar{x}_4) (x_2 + \bar{x}_3)$$

(b) $f^{FCD}(x_1, x_2, x_3, x_4) = \sum (1, 4, 7, 9, 11, 12, 13, 14)$
 $d = 5 = 6 = 13 = *$

x_3x_4 \ x_1x_2	00	01	11	10
00	0	1 ₄	0	0
01	1 ₁	d ₅	d ₁₃	1 ₉
11	0	1 ₇	0	0
10	0	d ₆	1 ₁₄	0

Annotations: $\bar{x}_1x_2$ (top row), $\bar{x}_3x_4$ (middle row), $x_2x_3\bar{x}_4$ (bottom row).

$f^{FMD}(x_1, x_2, x_3, x_4) = \bar{x}_1x_2 + \bar{x}_3x_4 + x_2x_3\bar{x}_4$

x_3x_4 \ x_1x_2	00	01	11	10
00	0	1 ₄	0	0
01	1 ₁	d ₅	d ₁₃	1 ₉
11	0	1 ₇	0	0
10	0	d ₆	1 ₁₄	0

Annotations: $x_1\bar{x}_3\bar{x}_4$ (top row), $\bar{x}_2\bar{x}_4$ (top row), $x_1x_2\bar{x}_3$ (middle row), $x_1x_3x_4$ (bottom row), $\bar{x}_2x_3$ (left column).

$\bar{f}^{FMD}(x_1, x_2, x_3, x_4) = \bar{x}_2\bar{x}_4 + \bar{x}_2x_3 + x_1x_3x_4$

$f^{FMC}(x_1, x_2, x_3, x_4) = \overline{\bar{x}_2\bar{x}_4 + \bar{x}_2x_3 + x_1x_3x_4}$
 $= \overline{\bar{x}_2\bar{x}_4} \cdot \overline{\bar{x}_2x_3} \cdot \overline{x_1x_3x_4}$
 $= (x_2 + x_4)(x_2 + \bar{x}_3)(\bar{x}_1 + \bar{x}_3 + \bar{x}_4)$

$$(a) f^{FCO}(x_1, x_2, x_3, x_4) = \sum (1, 4, 5, 6, 7, 9, 13, 14)$$

Metoda Quine - Mc Cluskey

- 1 $\bar{x}_1 \bar{x}_2 \bar{x}_3 x_4$
- 4 $\bar{x}_1 x_2 \bar{x}_3 \bar{x}_4$
- 5 $\bar{x}_1 x_2 \bar{x}_3 x_4$
- 6 $\bar{x}_1 x_2 x_3 \bar{x}_4$
- 7 $\bar{x}_1 x_2 x_3 x_4$
- 9 $x_1 \bar{x}_2 \bar{x}_3 x_4$
- 13 $x_1 x_2 \bar{x}_3 x_4$
- 14 $x_1 x_2 x_3 \bar{x}_4$

Grupa	Indicii ter. cuplati	Echiv. binari $x_1 x_2 x_3 x_4$	Impl. prim
1	1	0 0 0 1 ✓	
	4	0 1 0 0 ✓	
2	5	0 1 0 1 ✓	
	6	0 1 1 0 ✓	
	9	1 0 0 1 ✓	
3	7	0 1 1 1 ✓	
	13	1 1 0 1 ✓	
	14	1 1 1 0 ✓	

Grupa	Indicii ter. cuplati	Echiv. binari $x_1 x_2 x_3 x_4$	Impl. prim
1	1,5	0 - 0 1 ✓	
	1,9	- 0 0 1 ✓	
	4,5	0 1 0 - ✓	
	4,6	0 1 - 0 ✓	
2	5,7	0 1 - 1 ✓	
	5,13	- 1 0 1 ✓	
	6,7	0 1 1 - ✓	
	6,14	- 1 1 0 ✓	
	9,13	1 - 0 1 ✓	
			α

Grupuri	Indicii ter. cuplate	Echiv. binari $x_1 \ x_2 \ x_3 \ x_4$	Impl. primi
I	1, 5, 9, 13	- - 0 1	β
	1, 9, 5, 13	- - 0 1	β
	4, 5, 6, 7	0 1 - -	γ
	4, 6, 5, 7	0 1 - -	γ

Implicanți primi		Termeni canonici							
		1	4	5	6	7	9	13	14
iPE	α				*				*
iPE	β	*		*			*	*	
iPE	γ		*	*	*	*			

$$f^{PHD}(x_1, x_2, x_3, x_4) = x_2 x_3 \bar{x}_4 + \bar{x}_3 x_4 + \bar{x}_1 x_2$$